

Excelerate Internship

Week 2 Deliverable

Data Integration, Cleaning & Validation Report

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Report : Week 2 Deliverable Data Integration, Cleaning & Validation Report

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1. Introduction

This report documents the Week 2 data integration, cleaning, validation, and exploratory analysis performed on the Learner and Opportunity datasets. The objective was to integrate the datasets, evaluate data quality, validate the integration process, and prepare a reliable dataset for analytics.

2. Dataset Overview

2.1 Learner Dataset

Rows: 15,029

Columns: 256

Purpose: Contains learner applications, status, payment information, cohorts and related metadata.

Query:

```
SELECT *  
  
FROM learner_data_raw  
  
LIMIT 10;
```

Screenshot Placeholder:

Data Output Messages Notifications											
Showing rows: 1 to 10 Page No: 1 of 1											
pk	sk	assigned_cohort	fee	status	accept_reject_date	amount_to_be_paid	transaction_id	payment_status	Lagree_to_ter		
text	text	text	text	text	text	text	text	text	text		
1	Learner#	Learner#0000000010256C78JF7M6VQXQ2	[null]	0	[null]	[null]	[null]	[null]	[null]		
2	Learner#638ff3db-f3de-4ae4-9c06-29632039ad...	Learner#00000000001RVRMC22H9RYCY...	[null]	[null]	1010	[null]	0	[null]	unpaid		
3	Learner#	Learner#00000000010K1EA93ZGXA0SSNG	[null]	33	[null]	[null]	[null]	[null]	[null]		
4	Learner#e84ae65-34d3-440a-bf04-f877d3bc5c...	Learner#00000000010MXXW8GCFYE5S30...	BFSHUJCM	[null]	1070	1697094086113	0	[null]	unpaid		
5	Learner#9f52ded-a970-470d-830a-c13df19749...	Learner#000000000108MP2H74SMG8HW...	BU5NLYK	[null]	1070	1774502025544	0	[null]	unpaid		
6	Learner#6d4c12bb-211d-459f-8b2d-653095f9b...	Learner#00000000010V1FQTM6225F4HG...	[null]	[null]	1030	1714217388011	1	[null]	unpaid	TRUE	
7	Learner#6f39a383-aa51-4642-8995-61e49649f6...	Learner#000000000104G9QFHECKMCD2A...	[null]	[null]	1010	[null]	0	[null]	unpaid	TRUE	
8	Learner#c577426-abe7-4f04-a8db-92cff7ca86bd	Learner#000000000001RVRMC22H9RYCY...	[null]	[null]	1030	1682333589131	200	[null]	unpaid		
9	Learner#e84ae65-34d3-440a-bf04-f877d3bc5c...	Learner#00000000010CTBSXC407YQRNB...	889LYP	[null]	1070	1697088776119	0	[null]	unpaid		
10	Learner#0ca14518-2878-4135-adc1-06e6f0605...	Learner#00000000010VM9E97BOCBPMNZ...	B3V0FWZ	[null]	1120	1721382458168	0	[null]	unpaid		

Figure 2.1 - Learner Dataset Structure

2.2 Opportunity Dataset

Rows: 5,733

Columns: 33

Purpose: Contains opportunity details including opportunity_id, category, role, reward, duration and location.

Query:

```
SELECT *
```

LIMIT 10;

Data Output			Messages	Notifications
Showing rows: 1 to 10				
Page No: 1 of 1				
#	pk text	opportunity_id text	badge text	
1	Opportun...	Opportunity#000000000GBDAX3Z6VYRG7R...	(sk.Badge#00000000000G87VEXNYKKKTJKE9.ref_properties:(relation_id:\K9MSZJ))	
2	Opportun...	Opportunity#00000000000FKZAN696ZFW2V...	(sk.Badge#00000000000GAR6YSZ2B71HZJPM3.ref_properties:(relation_id:\6AJJ5D))	
3	Opportun...	Opportunity#00000000010ZFE0ZS22FA3Q62R4	NULL	
4	Opportun...	Opportunity#00000000000G34EPWZAJ6BDP...	(sk.Badge#00000000000GGQ1D2TSSHRS509.ref_properties:(relation_id:\74D0V7Q))	
5	Opportun...	Opportunity#00000000010MND1AYPPQZMEQ...	(sk.Badge#00000000010CASRZKZMTTQR06Q.created_at:1743161443422.ref_properties:(relation_id:\UOPXYI))	
6	Opportun...	Opportunity#00000000000QHDFWJ9R1HQJ4...	NULL	
7	Opportun...	Opportunity#00000000000QKBHNSCCDYAJ3T...	(sk.Badge#00000000000GKSA7GFSWAQ38BGV6.ref_properties:(relation_id:\5DTUK1)) (sk.Badge#000000001034MKCBKAWJYBAZV6.created_at:1710414867558.ref_properties:(relation_id:\WZAB2M))	
8	Opportun...	Opportunity#0000000010DZK12SW68Z6802E...	(sk.Badge#0000000001068CPTGB1Y2D0VQ.created_at:172770449137.ref_properties:(relation_id:\WZAB2M))	
9	Opportun...	Opportunity#00000000000GRHYKGM2HQFC6...	(sk.Badge#00000000000G1X85MGR4KRAZYK8.ref_properties:(test:test,relation_id:\9GL03J))	
10	Opportun...	Opportunity#00000000010YHJCSGAKECPBF...	NULL	

3. Data Integration Process

Join Logic

SQL Query

```
SELECT COUNT(*)  
FROM learner_data_raw l  
INNER JOIN opportunity_data_raw o  
ON REPLACE(l.sk,'Learner#','Opportunity#') =  
o.opportunity_id;
```

Screenshot Placeholder:


```

COUNT(*) - COUNT(sk) AS null_sk,

COUNT(*) - COUNT(status) AS null_status,

COUNT(*) - COUNT(assigned_cohort) AS null_assigned_cohort,

COUNT(*) - COUNT(opportunity_name) AS null_opportunity_name,

COUNT(*) - COUNT(opportunity_category) AS null_opportunity_category

FROM integrated_data;

```

Screenshot Placeholder:

Data Output Messages Notifications						
	total_rows bigint	null_sk bigint	null_status bigint	null_assigned_cohort bigint	null_opportunity_name bigint	null_opportunity_category bigint
1	14799	0	5749	9247	1	0

Figure 4.1 - NULL Analysis

Duplicate Analysis

Duplicate learner SK values were detected. These are considered valid because the dataset represents transactional/application records.

SQL Query

```

SELECT

sk,

COUNT(*)

FROM integrated_data

GROUP BY sk

HAVING COUNT(*) > 1

ORDER BY COUNT(*) DESC;

```

Screenshot Placeholder:

Data Output		Messages	Notifications
SQL			
	sk text		count bigint
1	Learner#000000001048MTP51YSZDRT0...		119
2	Learner#000000001034G8DGY0BAHV2Y...		69
3	Learner#000000000GZAETZA5Q0790HK...		43
4	Learner#000000000GEA0ACZBFEJARTB...		42
5	Learner#0000000010030P0X322MKT4P...		38
6	Learner#000000000GBD0AX3Z6VYRG7R...		36
7	Learner#0000000010HQ4WNAN1FSEVH...		32
8	Learner#000000000GT2180PM3XRQYRN...		32
9	Learner#0000000010WH2F20EF6AM7S9...		31
10	Learner#0000000010NK4VQ8MX97R828...		29
11	Learner#00000000101FD184RR5STPJCSV		28
12	Learner#000000000GZGGP7MHN0AX79...		25
13	Learner#000000000G8Z9ZW7B36BVHV...		25
14	Learner#0000000010PZWC1NRZRM2KM...		25
15	Learner#000000000GN3D9244GRCKPC7...		25
16	Learner#000000000GQK8HHSCCDYA3T...		25
17	Learner#000000000GBZ5VRTC3YS9T71...		24
18	Learner#000000000G1RYRMC22H9RYCY...		23
19	Learner#000000000G83249B39FZSAS08X		23
20	Learner#000000000GQHDFWJ9R1HQQJ4...		22
21	Learner#0000000010DSMQ4NDXQ7YRZ...		22
22	Learner#0000000010Q9H38929TXC2B05J		22
23	Learner#000000000GTPDXMC0Z1AJC6...		21
Total rows: 3173		Query complete 00:00:00.315	

Figure 4.2 - Duplicate Analysis

Unmatched Records

230 learner records did not join with the opportunity dataset.

SQL Query:

SELECT

```

CASE

    WHEN sk IS NULL THEN 'NULL SK'

    WHEN sk LIKE 'Learner#%' THEN 'Valid Learner ID'

    ELSE 'Invalid/Malformed SK'

END AS sk_status,

COUNT(*) AS total

FROM learner_data_raw l

LEFT JOIN opportunity_data_raw o

ON REPLACE(l.sk,'Learner#','Opportunity#') = o.opportunity_id

WHERE o.opportunity_id IS NULL

GROUP BY sk_status;

```

Screenshot Placeholder:

Data Output			Messages	Notifications
≡ 📄 ▼ 📋 ▼ 🗑️ 🔄 ⬇️ 📈 SQL				
	sk_status text	total bigint		
1	Invalid/Malformed ...	46		
2	NULL SK	184		

Figure 4.3 - Unmatched Records

5. Data Validation

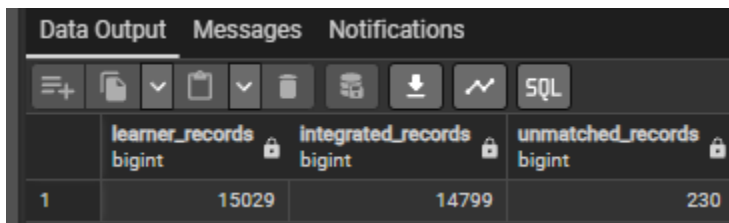
Validation Results

SQL Query :

SELECT

```
(SELECT COUNT(*) FROM learner_data_raw) AS learner_records,  
(SELECT COUNT(*) FROM integrated_data) AS integrated_records,  
(  
    SELECT COUNT(*)  
    FROM learner_data_raw l  
    LEFT JOIN opportunity_data_raw o  
    ON REPLACE(l.sk,'Learner#','Opportunity#') = o.opportunity_id  
    WHERE o.opportunity_id IS NULL  
) AS unmatched_records;
```

Screenshot Placeholder:



	learner_records bigint	integrated_records bigint	unmatched_records bigint
1	15029	14799	230

Figure 5.1 – Validation Summary

6. Exploratory Data Analysis

6.1 Opportunity Categories

SQL Query:

SELECT

```
    opportunity_category,
```

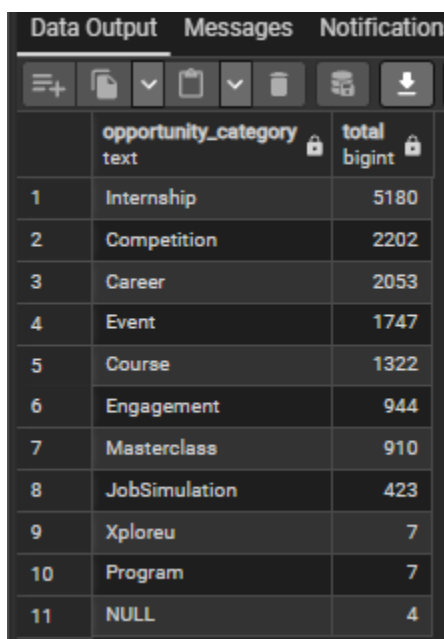
COUNT(*) AS total

FROM integrated_data

GROUP BY opportunity_category

ORDER BY total DESC;

Screenshot Placeholder:



The screenshot shows a data table with two columns: 'opportunity_category' (text) and 'total' (bigint). The table is sorted by 'total' in descending order. The categories and their counts are: Internship (5180), Competition (2202), Career (2053), Event (1747), Course (1322), Engagement (944), Masterclass (910), JobSimulation (423), Xploreu (7), Program (7), and NULL (4). The table is displayed in a dark-themed interface with a toolbar at the top.

	opportunity_category text	total bigint
1	Internship	5180
2	Competition	2202
3	Career	2053
4	Event	1747
5	Course	1322
6	Engagement	944
7	Masterclass	910
8	JobSimulation	423
9	Xploreu	7
10	Program	7
11	NULL	4

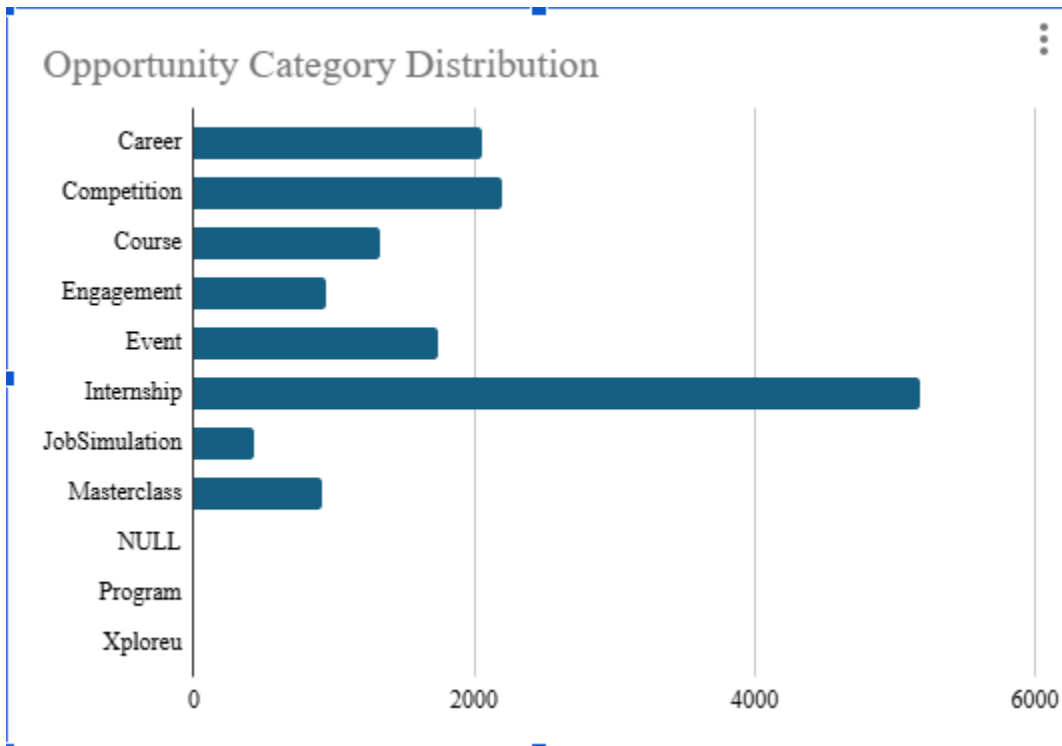
Observation:

- The integrated dataset contains opportunities across multiple categories.
- **Internship** has the highest number of records (5,180), followed by **Competition** (2,202) and **Career** (2,053).
- Categories such as **Program**, **Xploreu**, and **NULL** have very few records.

Insight:

- Internship opportunities dominate the dataset, indicating that the platform primarily focuses on internship-based programs.
- Categories with very low counts may represent specialized or less frequently offered opportunities.

Recommended Chart:



6.2 Learner Status

SQL Query:

SELECT

status,

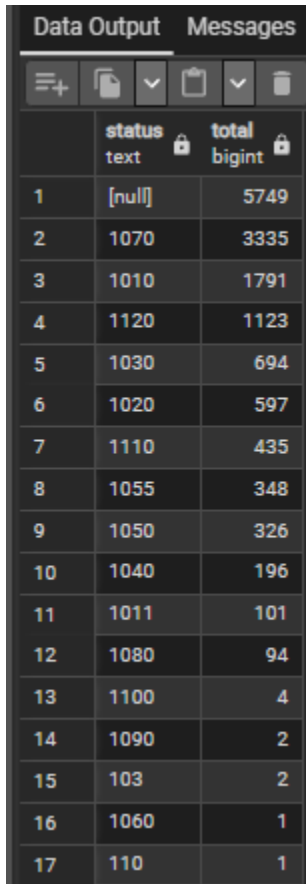
COUNT(*) AS total

FROM integrated_data

GROUP BY status

ORDER BY total DESC;

Screenshot Placeholder:



The screenshot shows a data table with two columns: 'status' (text) and 'total' (bigint). The table has 17 rows, numbered 1 to 17. The 'status' column contains various codes, and the 'total' column contains corresponding numerical values. The table is displayed in a dark-themed interface with a toolbar at the top.

	status text	total bigint
1	[null]	5749
2	1070	3335
3	1010	1791
4	1120	1123
5	1030	694
6	1020	597
7	1110	435
8	1055	348
9	1050	326
10	1040	196
11	1011	101
12	1080	94
13	1100	4
14	1090	2
15	103	2
16	1060	1
17	110	1

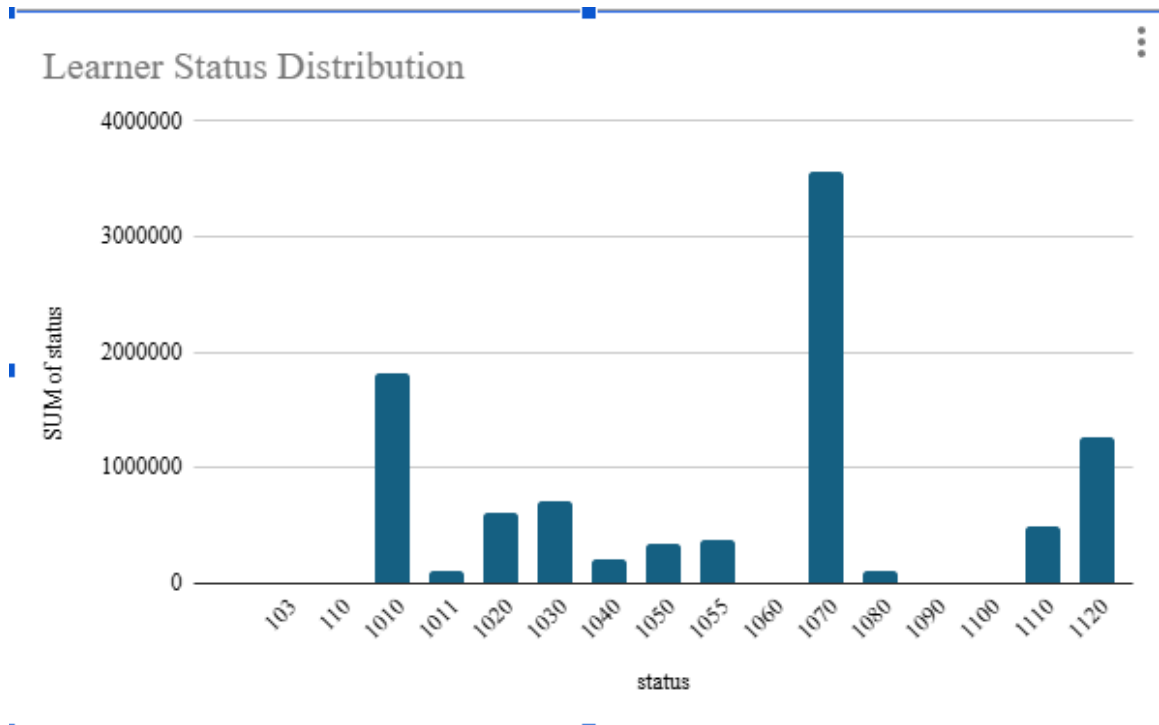
Observation:

- Learners are distributed across multiple application status codes (e.g., 1010, 1020, 1030, 1070, etc.).
- Some status codes contain significantly more records than others.

Insight:

- The learner application process consists of multiple workflow stages.
- The concentration of learners in a few status codes suggests that most applications are clustered within specific stages of the recruitment or approval process.

Recommended Chart:



6.3 Payment Status

SQL Query:

SELECT

payment_status,

COUNT(*) AS total

FROM integrated_data

GROUP BY payment_status

ORDER BY total DESC;

Screenshot Placeholder:

	payment_status	total
	text	bigint
1	unpaid	8529
2	[null]	5766
3	paid	504

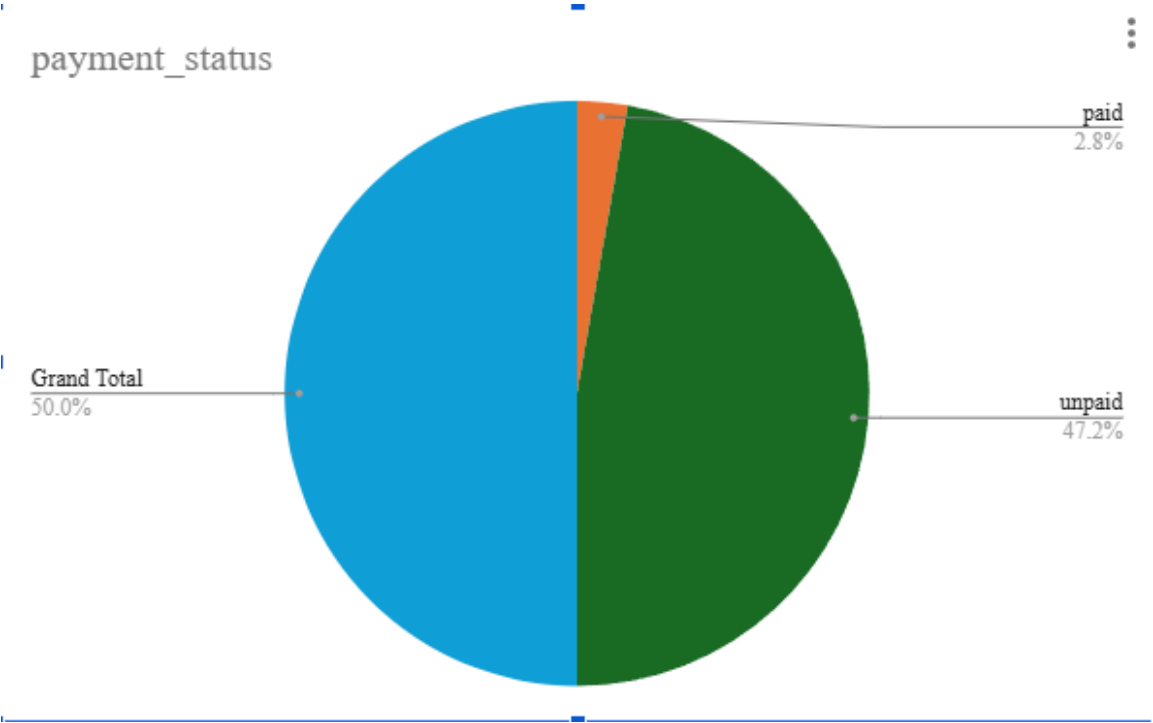
Observation:

The dataset contains different payment statuses, with **Unpaid** appearing most frequently.

Insight:

- A large proportion of learners have not completed the payment process, indicating that many applications are either pending or belong to free opportunities.

Recommended Chart:



6.4 Assigned Cohorts

SQL Query:

```
SELECT  
  
    assigned_cohort,  
  
    COUNT(*) AS total  
  
FROM integrated_data  
  
WHERE assigned_cohort IS NOT NULL  
  
GROUP BY assigned_cohort  
  
ORDER BY total DESC  
  
LIMIT 20;
```

Screenshot Placeholder:



The screenshot shows a data table interface with a dark theme. At the top, there are three tabs: 'Data Output', 'Messages', and 'Notifications'. Below the tabs is a toolbar with icons for menu, copy, paste, delete, download, and a 'SQL' button. The table has two columns: 'assigned_cohort' (text) and 'total' (bigint). The data is as follows:

	assigned_cohort text	total bigint
1	B9ZYF9D	68
2	B28EPFO	62
3	B360ZU8	32
4	BAD40TQ	31
5	BC5AWSN	30
6	B202074	22
7	B905078	20
8	BSQNZSD	19
9	BMSFPMW	18
10	BOXRCTU	17
11	B811759	14
12	B375816	14
13	BYTU1M9	12
14	BMP0KKY	12
15	B161629	12
16	BA9VXIL	12
17	BI983Q1	11
18	B526432	11
19	B495891	11
20	BIGLF6A	11

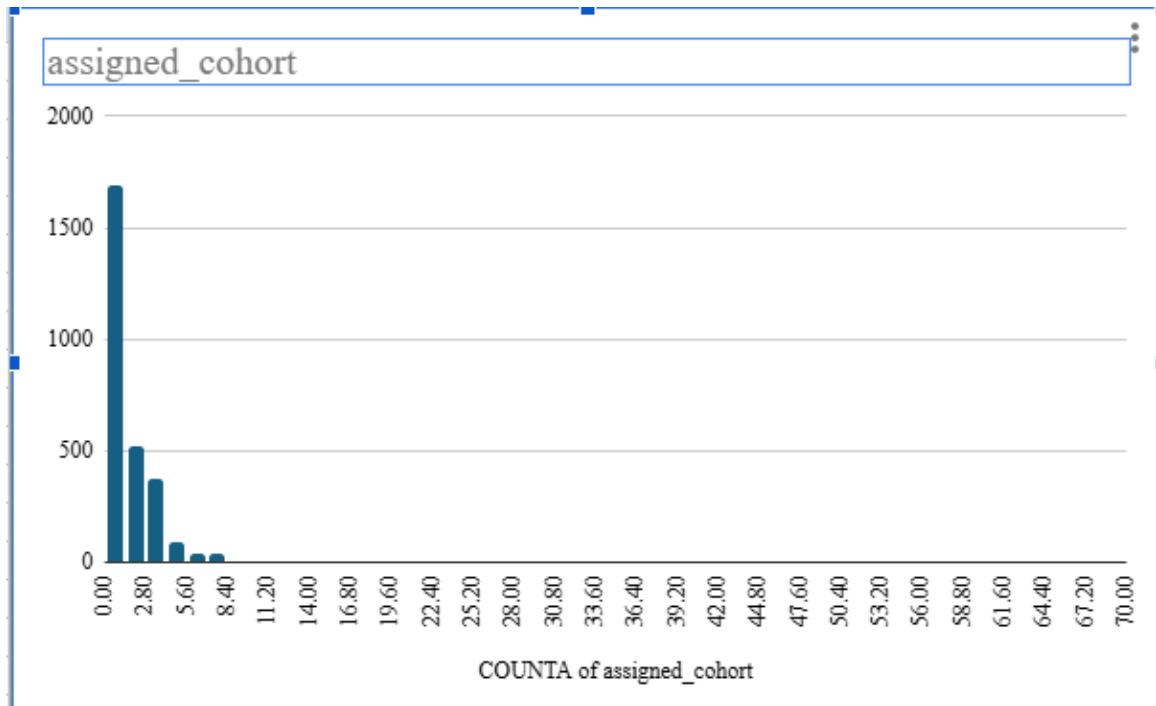
Observation:

Learners are assigned to multiple cohorts, with some cohorts having considerably higher enrollments than others.

Insight:

The uneven distribution of learners across cohorts indicates varying popularity or capacity among different training batches.

Recommended Chart:



6.5 Opportunity Locations

SQL Query:

SELECT

location,

COUNT(*) AS total

FROM integrated_data

WHERE location IS NOT NULL

GROUP BY location

ORDER BY total DESC

LIMIT 20;

Screenshot Placeholder:

Data Output			Messages	Notifications
≡+ 📄 ▼ 📋 ▼ 🗑 🔍 ⬇ 📈 SQL				
	location text	total bigint		
1	Virtual	2429		
2	null	1352		
3	virtual	1327		
4	WORK FROM HOME	1296		
5	Work From Home	1294		
6	NULL	1211		
7	wfm	1197		
8	vitruai	1185		
9	work from home	1165		
10	Null	1157		
11	WFM	1130		
12	Location	14		
13	Quia ad ipsum quos	11		
14	Commodi et aut sunt	10		
15	Opportunity name Page Testi...	7		
16	Quis voluptate totam	5		
17	Recusandae Ex volup	2		
18	Qui voluptatibus nat	2		
19	Elit cum quis volup	1		
20	Aliquam officia est	1		

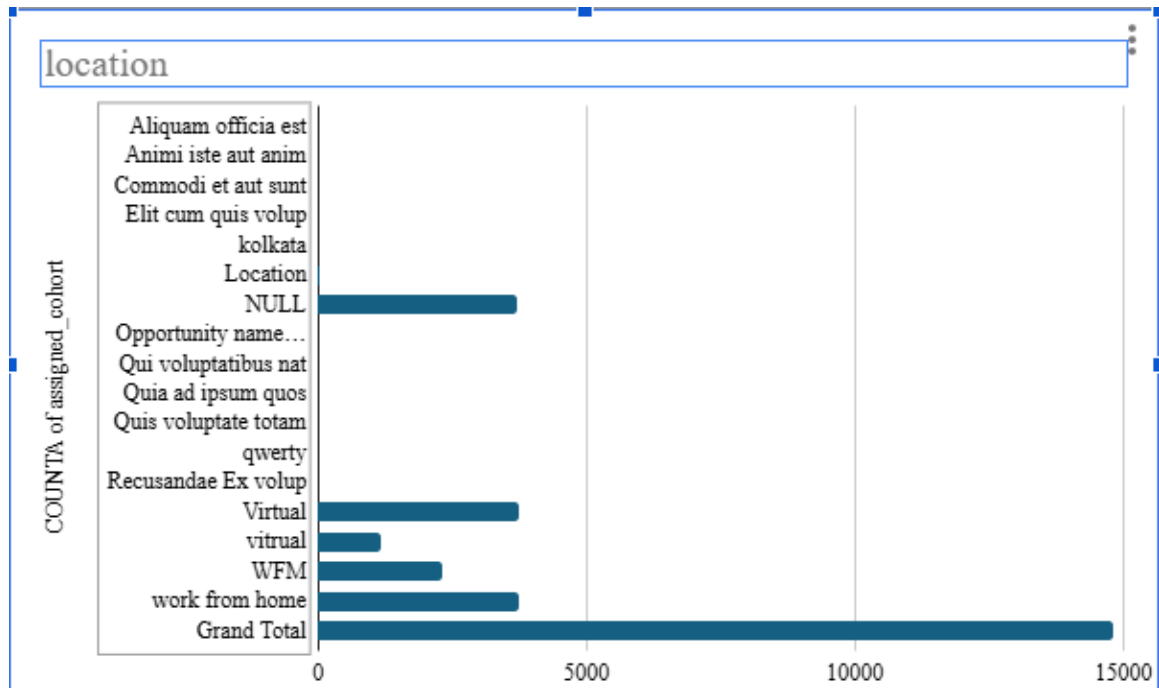
Observation:

- Opportunities are available across multiple locations.
- Certain locations have a noticeably higher number of opportunities.

Insight:

Opportunities are concentrated in selected locations, reflecting regional demand or organizational priorities.

Recommended Chart:



6.6 Opportunity Roles

SQL Query:

SELECT

role,

COUNT(*) AS total

FROM integrated_data

WHERE role IS NOT NULL

GROUP BY role

ORDER BY total DESC

LIMIT 20;

Screenshot Placeholder:

Data Output			Messages	Notification
	role	total		
	text	bigint		
1	NULL	8753		
2	Manager	1961		
3	Role	940		
4	Tester	257		
5	intern	187		
6	internship	143		
7	manager	127		
8	Social Media Analyst	123		
9	role	87		
10	Administrator	85		
11	Testing and debuggi...	84		
12	Intern	75		
13	tester	61		
14	Test	61		
15	test	35		
16	Instructor	35		
17	dfsdfdf	34		
18	Student	31		
19	ROle	27		
20	TesterHsa	27		

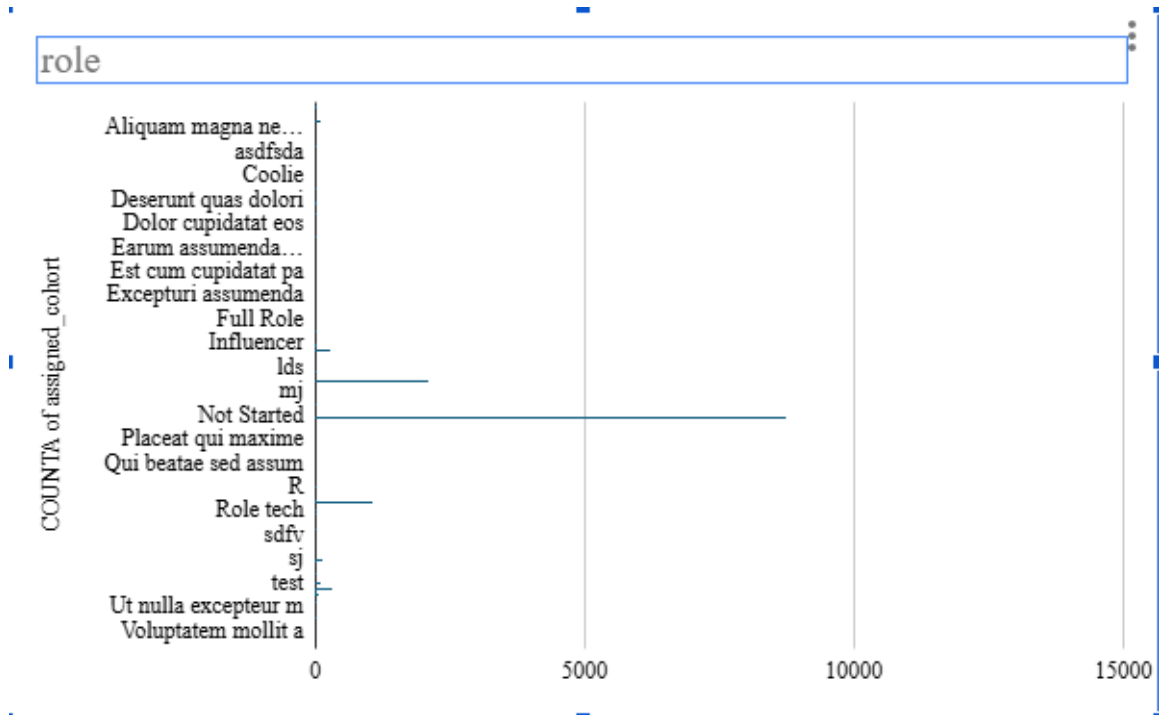
Observation:

- The dataset contains multiple job roles with varying frequencies.
- A few roles dominate the overall distribution.

Insight:

- The organization focuses on a limited number of high-demand roles while offering smaller numbers of specialized positions.

Recommended Chart:



6.7 Reward Distribution

SQL Query:

SELECT

reward,

COUNT(*) AS total

FROM integrated_data

WHERE reward IS NOT NULL

GROUP BY reward

ORDER BY reward;

Screenshot Placeholder:

Data Output			Messages	Notifications		
SQL			Showing rows: 1 to 1000		Page No: 1	of 6
	reward text	total bigint				
1	{ref_properties:{aaaaa}\relation_id\relation_id\RCY6UT\Satus\10100},ak-Reward#0000000000GP9G70RKA2V5WXNSB},{ak-Reward#0000000000GS48VZTYCVPK90CE1.created_at:1687949634116.modified_at:16...	23				
2	{ref_properties:{teat}\1234\relation_id\WARYU5},ak-Reward#0000000000GPWT65H92PVBCQ43},{ak-Reward#0000000000GK743JP3HHOP9SRB0.ref_properties:{relation_id\FCDLF8}}	1				
3	{ak-Reward#0000000000G26Q14QCAM1TH6X9C.created_at:1687414648737.ref_properties:{relation_id\Q9HC51}},(ak-Reward#000000000103XXBG9BK42QSSWN8.created_at:1687414648822.ref_properties:{relatio...	1				
4	{ak-Reward#0000000000G26Q14QCAM1TH6X9C.created_at:1687438960002.ref_properties:{relation_id\JMEHJ9}}	5				
5	{ak-Reward#0000000000G26Q14QCAM1TH6X9C.created_at:1695810252923.ref_properties:{relation_id\JF8W71E}},(ak-Reward#0000000000GP9G70RKA2V5WXNSB.created_at:1695810252933.ref_properties:{relatio...	2				
6	{ak-Reward#0000000000G26Q14QCAM1TH6X9C.created_at:1697528308750.ref_properties:{relation_id\J114HCS}}	3				
7	{ak-Reward#0000000000G26Q14QCAM1TH6X9C.created_at:1700549673236.ref_properties:{relation_id\B33LBD}}	1				
8	{ak-Reward#0000000000G26Q14QCAM1TH6X9C.created_at:1702454127816.ref_properties:{relation_id\KJUMOW}}	2				
9	{ak-Reward#0000000000G26Q14QCAM1TH6X9C.created_at:1704265660742.ref_properties:{relation_id\IH33VW9}}	2				
10	{ak-Reward#0000000000G26Q14QCAM1TH6X9C.created_at:1704352492818.ref_properties:{relation_id\CSPMQB}}	1				
11	{ak-Reward#0000000000G26Q14QCAM1TH6X9C.created_at:1704361256003.ref_properties:{relation_id\F5XANM}}	2				
12	{ak-Reward#0000000000G26Q14QCAM1TH6X9C.created_at:1705925318357.ref_properties:{relation_id\1F0331}}	3				
13	{ak-Reward#0000000000G26Q14QCAM1TH6X9C.created_at:1706011092037.ref_properties:{relation_id\1EL17Z5}}	3				
14	{ak-Reward#0000000000G26Q14QCAM1TH6X9C.created_at:1706012719136.ref_properties:{relation_id\15ZNLPA}}	3				
15	{ak-Reward#0000000000G26Q14QCAM1TH6X9C.created_at:1706014359147.ref_properties:{relation_id\1NIKOR7}}	3				
16	{ak-Reward#0000000000G26Q14QCAM1TH6X9C.created_at:1706607861146.ref_properties:{relation_id\1SFP29S}}	3				
17	{ak-Reward#0000000000G26Q14QCAM1TH6X9C.created_at:1706612741745.ref_properties:{relation_id\1AR796V}}	6				
18	{ak-Reward#0000000000G26Q14QCAM1TH6X9C.created_at:1706772506773.ref_properties:{relation_id\1DMNE8A}}	3				
19	{ak-Reward#0000000000G26Q14QCAM1TH6X9C.created_at:1707124906970.ref_properties:{relation_id\19ZWTAH}}	5				
20	{ak-Reward#0000000000G26Q14QCAM1TH6X9C.created_at:1708943695086.ref_properties:{relation_id\1HFPQS}}	1				
21	{ak-Reward#0000000000G26Q14QCAM1TH6X9C.created_at:1708944908242.ref_properties:{relation_id\19BHOP7}}	1				
22	{ak-Reward#0000000000G26Q14QCAM1TH6X9C.created_at:1709023201567.ref_properties:{relation_id\1GLO1XM}}	1				
23	{ak-Reward#0000000000G26Q14QCAM1TH6X9C.created_at:1709025333138.ref_properties:{relation_id\13QVK7S}}	1				
Total rows: 5363 Query complete 00:00:00.294						
CRLF Ln 177, Col 1						

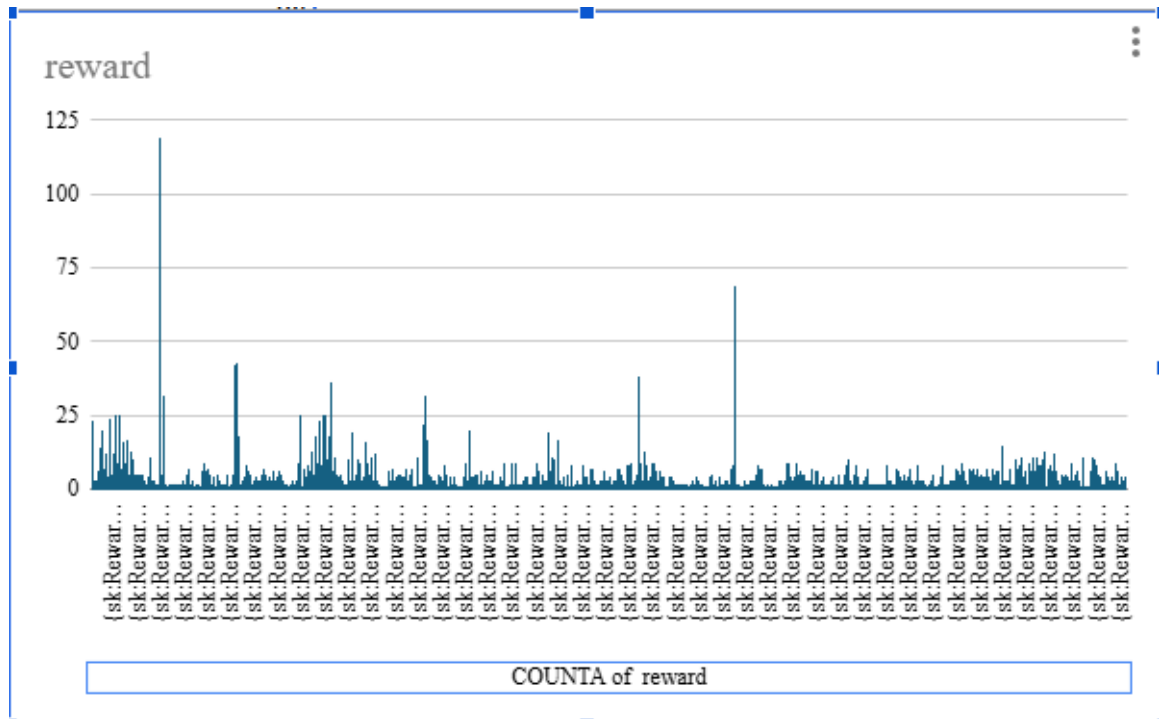
Observation:

- Reward values vary across different opportunities.
- Some reward amounts occur much more frequently than others.

Insight:

- Standardized reward structures appear to be used for many opportunities, while higher-value rewards are limited to selected programs.

Recommended Chart:



6.8 Duration Analysis

SQL Query:

SELECT

duration_type,

ROUND(AVG(duration::NUMERIC),2) AS avg_duration,

MIN(duration::NUMERIC) AS min_duration,

MAX(duration::NUMERIC) AS max_duration

FROM integrated_data

WHERE duration ~ '^[0-9]+(\.[0-9]+)?\$'

GROUP BY duration_type

ORDER BY duration_type;

Screenshot Placeholder:



The screenshot shows a data table with the following columns: duration_type (text), avg_duration (numeric), min_duration (numeric), and max_duration (numeric). The table is sorted by duration_type. The data is as follows:

	duration_type	avg_duration	min_duration	max_duration
1	da	1.00	1	1
2	day	1.52	1	6
3	days	308.90	2	11334
4	hour	1.08	0	3
5	hours	19.20	2	200
6	minute	1.00	1	1
7	minutes	218.97	2	10000
8	month	1.40	1	22
9	months	5.32	2	120
10	NULL	58.00	58	58
11	week	1.05	0	4
12	weeks	30.90	1	3030
13	year	1.00	1	1
14	years	18.29	1	60
15	yearssss	2.00	2	2

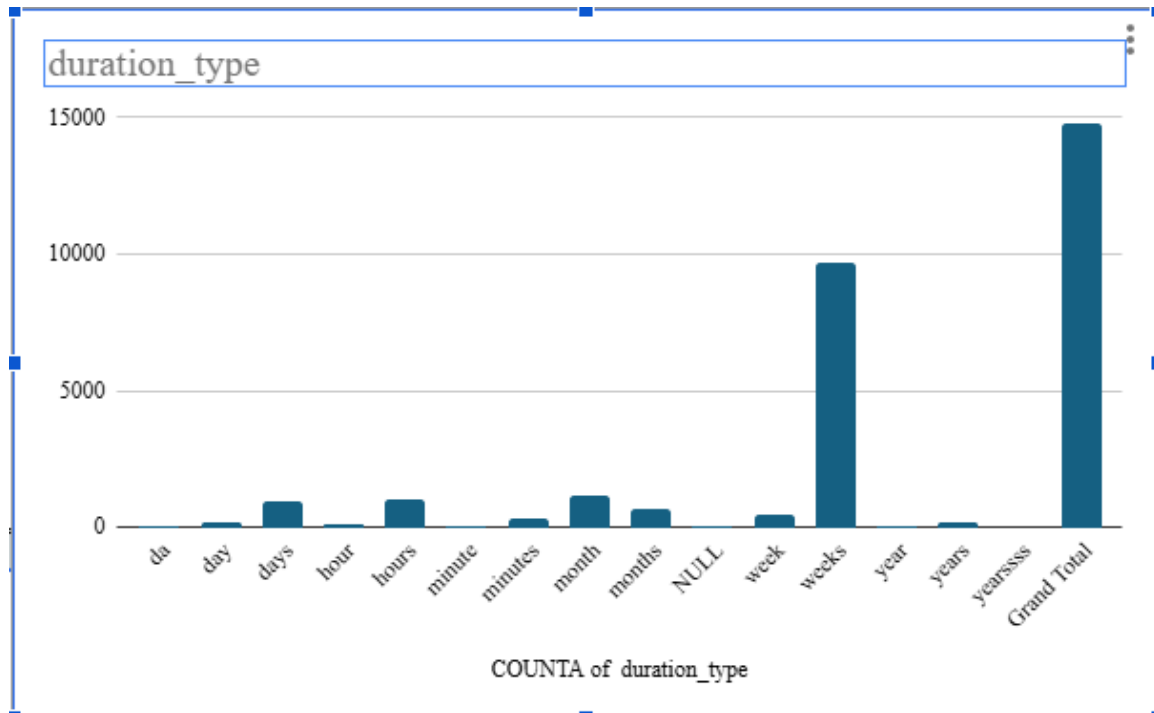
Observation:

- Opportunities have different duration types (e.g., days, weeks, months).
- One duration type is more common than the others.

Insight:

Most opportunities follow a standardized duration format, making program scheduling more consistent.

Recommended Chart:



6.9 Category vs Status

SQL Query:

SELECT

opportunity_category,

status,

COUNT(*) AS total

FROM integrated_data

GROUP BY opportunity_category, status

ORDER BY opportunity_category, total DESC;

Screenshot Placeholder:

Data Output				Messages	Notifications
	opportunity_category text	status text	total bigint		
1	Career	[null]	1088		
2	Career	1070	368		
3	Career	1010	209		
4	Career	1030	112		
5	Career	1120	103		
6	Career	1020	44		
7	Career	1110	43		
8	Career	1050	40		
9	Career	1055	29		
10	Career	1040	17		
11	Competition	[null]	738		
12	Competition	1010	474		
13	Competition	1070	450		
14	Competition	1011	98		
15	Competition	1120	98		
16	Competition	1030	90		
17	Competition	1020	74		
18	Competition	1110	60		
19	Competition	1055	47		
20	Competition	1050	43		
21	Competition	1040	30		
22	Course	[null]	632		
23	Course	1070	211		
Total rows: 101		Query complete 00:00:00.192			

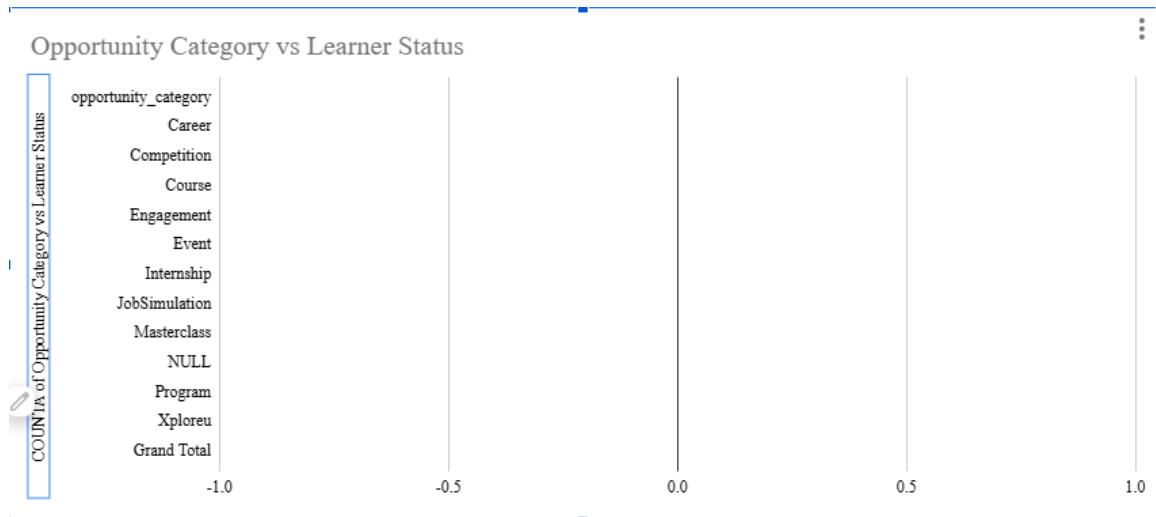
Observation:

- Learner status varies across different opportunity categories.
- Internship opportunities contain the largest number of learner records across multiple status codes.

Insight:

Internship programs receive the highest learner participation and exhibit activity across all stages of the application workflow, demonstrating their importance within the integrated dataset.

Recommended Chart:



7. Key Findings

- Successfully integrated 14,799 learner records.
- 230 records were unmatched (184 NULL sk, 46 malformed sk).
- Join logic validated using REPLACE(sk,'Learner#','Opportunity#').
- Integrated dataset is ready for visualization and reporting.

8. Conclusion

The objectives of Week 2 were achieved through successful data integration, data quality assessment, validation, and preparation of an integrated dataset for future dashboards and analytics.

Appendix A - SQL Queries

- INNER JOIN using REPLACE()
- NULL analysis

- Duplicate analysis
- Unmatched record analysis
- EDA Queries 1-9